

can ask whether models reflect the varied perspectives held by real human subpopulations, and if they do so (consistently) across a range of topics. We find that these two perspectives can often lead to different conclusions—e.g., `text-davinci-003` while very pro-liberal based on the modal view, does not capture liberal viewpoints in a nuanced and consistent manner according to our study.

Subjectivity in evaluations. There has been a long-standing push within the NLP community to consider the subjective and affective dimensions of language in evaluating models (Alm, 2011). Prior works show that for many tasks—from toxicity detection (Gordon et al., 2021, 2022; Davani et al., 2022; Sap et al., 2022; Goyal et al., 2022), ethics judgements (Lourie et al., 2021), and inference (Pavlick & Kwiatkowski, 2019)—there is inherent variability in what different humans consider the “correct answer”. These studies serve as a motivation for our work, where we approach the problem of evaluating opinions expressed by LMs through the use of surveys.

Human-LM alignment. There is a growing body of work seeking to make LMs more human-aligned (Askell et al., 2021; Ouyang et al., 2022; Glaese et al., 2022; Bai et al., 2022). While these works recognize the subjectivity of the alignment problem, they do not focus on it—seeking instead to identify values to encode in models and building techniques to do so. Our work looks instead delves deeper into the issue of subjectivity, asking *who* are the humans that we are/should be aligning the models to?

Bias, toxicity, and truthfulness. There is a long line of work studying the bias and fairness of NLP systems (Nadeem et al., 2020; Dhamala et al., 2021; De-Arteaga et al., 2019; Brown et al., 2020; Gao et al., 2021; Srivastava et al., 2022; Liang et al., 2022; Xu et al., 2021; Perez et al., 2022a; Ganguli et al., 2022). These works study properties of LMs such as bias, toxicity, and truthfulness, focusing on flagging undesirable outcomes when the gold standard behavior is somewhat well-defined (e.g., don’t use slurs). Our work takes a complementary perspective: evaluating LMs on inherently subjective questions taken from Pew Research. This allows us to gain quantitative insights into the representativeness of opinions expressed by LMs on contentious but important topics such as religion or privacy.

6 Conclusion

Our work puts forth a framework to examine the opinions reflected by LMs through the lens of public opinion polls. Using our OpinionQA dataset, we identify a number of ways in which LMs are not well-aligned with human opinions, including overall representativeness with respect to people in the US; subgroup representativeness on groups such as 65+, Mormon, and widowed; and steerability. Our work also contributes to the broader discourse around LMs, including questions of whether instruct-tuning distorts opinion distributions, and whether models hold consistent biases towards liberal views.

7 Limitations

While our work provides a quantitative lens into LM opinions, it suffers from three key limitations discussed below.

Limitations of alignment. Our approach analyzes LM opinions through the lens of *who* they align with. This approach allows us to precisely define our metrics and collect data, but also warrants caution – LMs that perfectly represent human opinions may not necessarily be desirable as they may also, in the process, replicate human biases. We view our metrics as useful ways to understand the behavior of LMs, and to allow model developers to identify when their models misrepresent specific groups, and not necessarily as benchmarks that should be blindly optimized.

Limitations of the ATP and surveys. While our methodology to build an evaluation dataset can be easily adapted to any multiple-choice survey, the OpinionQA dataset we instantiate is based on the American Trends Panel. Surveys in general may be sensitive to details such as question specificity (Berinsky, 2017) and the ATP in particular has had past issues with social desirability bias (Yan, 2021) that may affect the accuracy of the human opinion distribution. Beyond that, the ATP survey targets individuals in the US, making our conclusions valid only for the populations in the US. Many societies differ from WEIRD (Western, Educated, Industrialized, Rich and Democratic) societies such as the United States (Henrich et al., 2010) and there is a need for future work on global equivalents to OpinionQA.

Limitations of the multiple-choice format. Our work focuses on probing LM behaviors using a multiple-choice prompt taken from public opinion surveys. While the multiple-choice format allows us to precisely evaluate the models, it also differs from the open-ended text generation setting in which LMs are being increasingly used. It is an open question whether opinion alignment that is measured through multiple choice will be reflected in the downstream use cases of LMs – for example, will the liberal-leaning opinion alignment of RLHF fine-tuned models appear in a dialogue context or open-ended QA? Some recent works suggest that the group-alignment effects (e.g. to liberals) do reflect in other settings (Perez et al., 2022b; Hartmann et al., 2023), but whether these results transfer broadly warrants further investigation.

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